

Which Page Replacement Algorithm is Best?

Theoretically, the Optimal Page Replacement Algorithm is the best because it always produces the minimum number of page faults. However, practically, LRU (Least Recently Used) is considered the best for the following reasons:

1. Practical Implementation (Most Important)

The Optimal algorithm requires knowledge of future page references, which is impossible in practice. LRU uses past memory access history, making it practical for real operating systems.

2. No Belady's Anomaly

Normally, increasing the number of frames reduces page faults. However, in FIFO, some reference strings may produce more page faults even after increasing the number of frames (Belady's Anomaly). LRU avoids this problem, so page faults only decrease or remain the same.

3. Lowest Page Fault Rate

LRU generates page faults very close to the Optimal algorithm, resulting in fewer disk accesses and better system performance.

4. Better Frame Utilization

LRU keeps recently used pages in memory and replaces the least recently used page, making efficient use of available memory frames.